

following equation implicitly describes the ball's angle *theta* at each moment of its flight:

$$\tan(\theta) = \frac{10x - 5x^2}{2 - x} = 5x$$

The overall situation is complex: a ball is hit far away and quickly shoots through a parabolic curve whose end is hard to immediately estimate. But in this complex situation, Chapman has found this simple relationship: that *when the outfielder is standing in the right location*, the tangent of theta grows at a simple, constant rate. The kernel of Chapman's breakthrough is that the tangent of theta, the ball's angle with the ground, grows linearly over time. Since Chapman found that simple relationship in the weeds of the outfielder problem, he was able to develop an elegant algorithmic solution to it.

His solution depends on the fact that if something—in this case, the tangent of theta—grows at a constant rate, it has zero acceleration. So if you are standing exactly where a ball is headed, you'll observe an angle whose tangent experiences zero acceleration. By contrast, if you are standing too close to the ball's initial position, you'll observe positive acceleration. If you are standing too far from the ball's initial position, you'll observe negative acceleration. (You are encouraged to verify the messy calculus behind these truths if you so desire.) This means that an outfielder can know where they need to go by feeling how steadily they have to tilt back their head as they look at the ball rising—thinking, so to speak, with their neck.

Applying Chapman's Algorithm

Robots don't necessarily have necks, and so a method for "thinking with one's neck" may not be helpful for a robot outfielder. Remember that they can solve quadratic equations directly and instantaneously to find where to go to catch a ball, without worrying about the acceleration of the tangent of theta. But for humans, Chapman's neck-thinking method could be extremely useful. In order to get to the ball's eventual destination, a human outfielder could follow this relatively simple process: